

StateVLM: A State-Aware Vision-Language Model for Robotic Affordance Reasoning

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Abstract

Vision-language models (VLMs) have shown remarkable performance in various robotic tasks, as they can perceive visual information and understand natural language instructions. However, when applied to robotics, VLMs remain subject to a fundamental limitation inherent in large language models (LLMs): they struggle with numerical reasoning, particularly in object detection and object-state localization. To explore numerical reasoning as a regression task in VLMs, we propose a novel training strategy to adapt VLMs for object detection and object-state localization. This approach leverages box decoder outputs to compute an Auxiliary Regression Loss (ARL) during fine-tuning, while preserving standard sequence prediction at inference. We leverage this training strategy to develop StateVLM (State-aware Vision-Language Model), a novel model designed to perceive and learn fine-grained object representations, including precise localization of objects and their states, as well as graspable regions. Due to the lack of a benchmark for object-state affordance reasoning, we introduce an open-source benchmark, Object State Affordance Reasoning (OSAR), which contains 1,172 scenes with 7,746 individual objects and corresponding bounding boxes. Comparative experiments on adapted benchmarks (RefCOCO, RefCOCO+, and

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RefCOCOg) demonstrate that ARL improves model performance by an average of 1.6% compared to models without ARL. Experiments on the OSAR benchmark further support this finding, showing that StateVLM with ARL achieves an average of 5.2% higher performance than models without ARL. In particular, ARL is also important for the complex task of affordance reasoning in OSAR, where it enhances the consistency of model outputs.

Keywords: Vision-language models, Referring expression comprehension, Visual perception, Object-state understanding, Affordance reasoning

1. Introduction

Vision-language models (VLMs) [1, 2] for robotics are not only capable of natural language understanding but also of visual perception. In robotic tasks such as manipulation, human-robot interaction, and autonomous driving, precise localization and understanding of objects within a scene are essential. However, VLMs applied to robotics remain subject to a fundamental limitation inherent in large language models (LLMs): they struggle with numerical reasoning, particularly object detection and object-state localization [3, 4, 5].

Most important VLMs architectures adopt LLMs as their backbone and integrate visual perception modules within them [6, 7, 8, 9, 10, 11, 12, 13, 14]. LLMs are originally trained on sequence prediction. The existing VLMs focused on object detection tasks use a sequence-based output format, which is the LLMs’ default. For instance, Pix2Seq [15] is an object detection framework that predicts a sequence of discrete tokens that correspond to object descriptions (e.g., object bounding boxes and class labels, $y_{\min} = 9$ $x_{\min} = 7$ $y_{\max} = 67$ $x_{\max} = 98$ `train.....`). State-of-the-art VLMs also formulate object detection as a sequence prediction task, including SPHINX [6], Shikra [7], LLAVA-G [10], and others. However, such a representation is sub-optimal because discrete symbolic tokens require sequence modeling, whereas continuous numerical outputs are better modeled with regression. **We hypothesize that using this Pix2Seq training paradigm for numerical tasks may result in inefficient learning.**

To explore object detection as a regression task in VLMs, Zhang et al. proposed the Pix2Emb method, NExT-Chat, which introduces a box decoder for continuous numerical outputs as embeddings [16]. The key distinction between Pix2Seq and Pix2Emb methods lies in their outputs: Pix2Seq

method generates sequences only, whereas Pix2Emb method generates both sequences and embeddings. However, despite adopting a similar training data regime for detection training, NExT-Chat performs slightly worse than the Pix2Seq method, Shikra-7B [7], on referring expression comprehension (REC). Zhang et al. propose two hypotheses to explain this gap. First, achieving an optimal balance between sequence and embedding losses is challenging, whereas Pix2Seq methods do not encounter this issue, as they optimize only the sequence loss. Second, LLMs are not pre-trained on regression tasks, which may further increase training difficulty. To further investigate this open question in VLMs, we propose a less intrusive strategy than Pix2Emb method for adapting VLMs to object detection and object-state localization.

Previously, there have been some benchmarks that focused on object-state recognition and classification [17, 18, 19, 20] and fewer studies on object-state localization [21, 22]. However, object-state localization is essential for robotic tasks such as object manipulation, which require affordance reasoning at the object-state level [23, 24]. State-level reasoning is significantly more challenging than category-level reasoning because object states are dynamic and can change over time [19, 25, 26]. For instance, two dirty plates may require different grasping areas depending on the specific locations of the dirt. Similarly, a knife can be entirely clean or dirty, or its handle and blade may exhibit different cleanliness conditions. In some causal scenarios, if a knife is accidentally dropped into food, it may have a clean blade but a dirty handle. In this special case, the knife’s blade becomes the appropriate grasping area. Therefore, to address the lack of a benchmark for object-state affordance reasoning, we introduce an open-source benchmark that focuses on object-state localization and affordance reasoning.

Our primary contribution in this paper is summarized as follows:

1. We propose StateVLM, a novel model designed to perceive and learn fine-grained object representations, including precise localization of objects and their states, as well as graspable regions.
2. To further investigate object detection as a regression task in VLMs, we propose a novel strategy to adapt VLMs to object detection. The model uses a box decoder output to compute an Auxiliary Regression Loss (ARL) during training (see Fig. 1a), while during inference it proceeds with standard sequence prediction (see Fig. 1b).
3. Comparative experiments on StateVLM demonstrate that this ARL

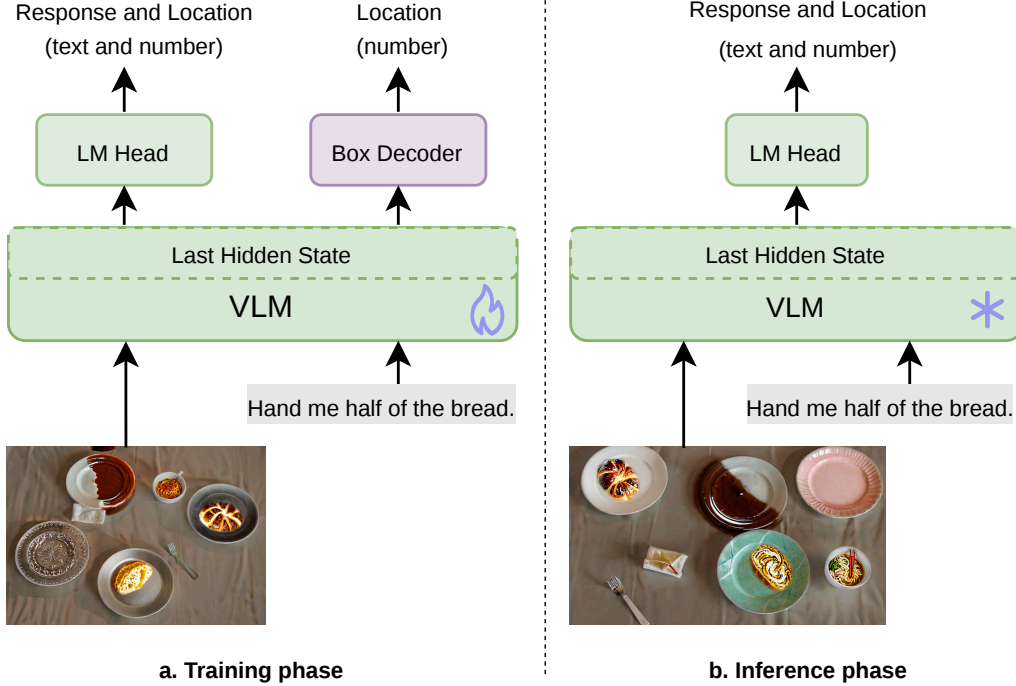


Figure 1: The overall framework of **StateVLM**. During the **training phase** (a), a specially designed box decoder converts sequence predictions into embedding predictions for calculating an auxiliary regression loss. In the **inference phase** (b), however, the model continues to rely on standard sequence-based text and number predictions, because the LLM backbone of the VLM is originally trained on sequence prediction.

significantly improves the StateVLM’s convergence during fine-tuning, thereby enhancing its ability to learn object location features. The experiments were conducted on the adapted referring expression comprehension (REC) task. For this task, we used the following datasets: RefCOCO, RefCOCO+, and RefCOCOg.

4. To address the lack of a benchmark for object-state affordance reasoning, we introduce an open-source benchmark OSAR to the research community, focused on object-state localization and affordance reasoning. It comprises both complex and simple scenes, totaling 1,172 scenes, and contains 7,746 individual objects, 25,401 referring expressions, and corresponding bounding boxes. LoRA fine-tuning of StateVLM on the OSAR benchmark further supports the effectiveness of ARL in improv-

ing the model’s state-aware capabilities.

The remainder of this paper is structured as follows. Section 2 reviews existing approaches to tackle object-state understanding and reasoning and highlights the limitations of current methods. Section 3 describes the task definition and provides a summary of our proposed benchmark, named OSAR. The architecture and training procedure of StateVLM are detailed in Section 4. Section 5 presents and analyzes the experimental results, then discusses our findings, highlighting the role of the auxiliary regression loss in improving performance. Finally, Section 6 concludes the paper and suggests directions for future work.

2. Related Work

2.1. VLMs for Referring Expression Comprehension

REC (Referring Expression Comprehension) is a fundamental region-level image understanding task that seeks to identify and localize a target object described by a natural language expression within a given scene [4, 5]. In recent years, State-of-the-art VLMs have achieved remarkable performance on this task [6, 7, 8, 9, 10, 11, 12, 13, 14], significantly surpassing traditional non-VLM approaches [27, 28, 29, 30, 31, 32]. This improvement can largely be attributed to the integration of visual representations or embeddings into LLMs. Due to their training on massive and diverse text corpora, LLMs exhibit strong linguistic interpretation abilities and extensive commonsense reasoning, both of which are crucial for advancing embodied AI and robotic perception. Nevertheless, these models demand substantial computational resources and prolonged multi-stage training processes, which present notable practical challenges.

The computational resources and training times of the most representative VLMs for the REC task are different. Depending on the number of transformer layers in the LLM and the scale of the visual backbone, they typically contain either 7B(illion) or 13B parameters. Furthermore, the models adopt several different training configurations. First, they are trained on computational clusters of varying sizes, such as 8, 32, or 256×A100 GPUs. Second, they employ various training strategies, including one-stage, two-stage, or multi-stage training. Third, training durations are reported inconsistently, with some measured in wall-clock time and others in training steps.

Therefore, it is difficult to draw a rigorous scientific conclusion because we lack controlled variables [33]. It remains unclear whether simply combining

two distinct data distributions (discrete text and continuous location data) results in inefficient training for bounding box coordinate prediction.

2.2. *Affordance Reasoning: From Object Categories to Properties and States*

Object affordance refers to the range of actions an agent can perform with an object, as perceived through its properties [24, 34]. A robust understanding of object categories (e.g., ‘a cup’, ‘a plate’, and ‘an apple’) provides a foundational basis for such reasoning. Building on the object categories, a model can further infer an object’s potential affordances by leveraging finer-grained information, object attributes.

Object attributes, as generalizable properties, are central to this reasoning process. For instance, instead of relying solely on category recognition, Yang et al. [26] developed a robotic grasping method based directly on object attributes. Similarly, Attr-POMDP [35] presents an attribute-guided formulation of a partially observable Markov decision process for task disambiguation. In the Octopi system [36], the authors selected hardness, roughness, and bumpiness as key physical attributes for physical reasoning. Another prominent direction in affordance reasoning, particularly for robotic manipulation, involves physically grounded methods [37, 38, 39]. PhyGrasp [40] is designed to accurately assess the physical properties of object parts to determine optimal grasping poses. This finer-grained information is complex, and some of its values are dynamic for any given object.

We ground the concept of an object state in object-oriented programming [41], where it is defined as follows: “The state of an object encompasses all of the (usually static) properties of the object, plus the current (usually dynamic) values of each of these properties.” Therefore, in our context, the properties refer to object properties (usually static), such as color, shape, size, weight, texture, cleanliness, and rigidity. The values of these properties are dynamic, meaning they can change over time or remain constant, and different properties may change at different rates. Our method centers on understanding an object’s state and reasoning about its affordances, which has been less explored so far.

2.3. *The Missing Link Between Object-State and Manipulation*

Previous benchmarks have primarily focused on object-state recognition and classification [17]. For example, the cooking state recognition challenge [18] focused on 18 types of objects and their corresponding states, such as diced, grated, or creamy paste. Another dataset [19] classified whether a

container was full or half empty. Furthermore, the Object State Detection Dataset [20] includes a larger number of objects and state classes.

Another line of the dataset focuses on detecting or anticipating object state changes. For example, a subset of Ego4D [25] addresses object state change classification and detection. Ego4D-OSCA [42] is a dataset for anticipating object state changes from Ego4D video sequences. The OSCaR dataset [22] focuses on object state captioning and state change representation. State-aware Keypoint Trajectories (SKT) [21] provides a synthetic dataset encompassing a broad spectrum of garment configurations, ranging from flat to deformed and folded states. However, none of these datasets explicitly focus on how an object’s state influences its manipulation.

Overall, further investigation is needed to determine whether simply combining discrete and continuous data distributions leads to inefficient training for bounding-box coordinate prediction in VLMs. Designing a multimodal dataset is essential for analyzing this hypothesis at the object-state level, especially for affordance-based reasoning for object manipulation.

3. Multimodal Dataset for Object State Affordance Reasoning

Robotic manipulation can be decomposed into macro- and micro-level tasks. The macro-level task involves task planning, where the robot identifies objects and reasons about their destinations. The micro-level task involves motion planning for grasping and placing the object, as shown in Fig. 2. Our previous study, OSSA [43], focused on the macro-level task of robotic task planning. However, this study did not address object-state localization and affordance reasoning, both of which are crucial for robotic manipulation. To investigate this missing aspect, we propose and make available a novel benchmark, **Object State Affordance Reasoning (OSAR)**.

3.1. Task Definition

Object Detection: Object-State Localization. Derived from the RefCOCO [4] dataset, we construct a new referring comprehensive expression dataset. The expressions are explicit descriptions of the target that relate to the object’s state and spatial location. The format is shown in Table 1.

Affordance Reasoning: Object Grasp Prediction. In contrast to object detection, object grasp prediction requires consideration of physical common sense. For instance, in everyday kitchen scenarios, references are typically

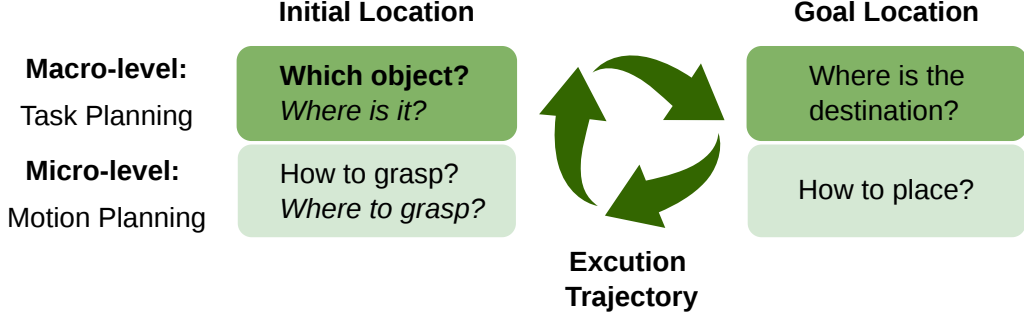


Figure 2: The decomposition of robotic manipulation into macro- and micro-level tasks. The macro-level task involves task planning, identifying objects, and their destinations. The micro-level task involves motion planning, grasping, and placing the objects. While an object’s state can affect all steps, we focus on its impact on localization, specifically determining the target object, its position, and the grasping location.

made to the object itself rather than to its container. During physical interaction, the decision of whether to grasp a container or its contents depends on the properties and state of the contained item. Specifically, one would grasp a container (e.g., a bowl) to pass noodles, whereas passing an apple does not require a container. Accordingly, we introduce the second task, termed object grasp prediction. Whose format is identical to that of object detection (see Table 1 for affordance reasoning), except that the predicted bounding box corresponds to the region intended for grasping.

3.2. Benchmark Dataset

3.2.1. Visual Scenes

We use the Stable Diffusion model [44] to generate images and render them in Blender¹. In total, we selected 1,172 images based on quality, comprising 780, 267, and 125 images for templates (a), (b), and (c), respectively, as illustrated in Fig. 3. From each scene, we selected 11.2% of the images as test samples. The final dataset comprises 7,746 individual objects, of which 11.17% belong to the test set. These objects were in different states within their respective categories, as shown in Fig. 4.

¹<https://www.blender.org/>

Table 1: Dialog templates in OSAR. The placeholder [target] is instantiated with the state of objects present in the corresponding scenes, such as ‘empty plate’, ‘dirty plate’, or ‘bowl with noodles’. The bounding box coordinates $\langle \text{box} \rangle x_1, y_1, x_2, y_2 \langle / \text{box} \rangle$ denote the object location in the object detection task and the grasp location in the affordance reasoning task, where (x_1, y_1) corresponds to the top-left corner and (x_2, y_2) corresponds to the bottom-right corner.

Task Types	Conversations	
	Roles	Content
Object detection (object-state localization)	User:	“Show me the [target].”
	StateVLM:	“response: Here is the [target]. $\langle \text{box} \rangle x_1, y_1, x_2, y_2 \langle / \text{box} \rangle$.”
	User:	“Where is the [target]?”
	StateVLM:	“response: Here is the [target]. $\langle \text{box} \rangle x_1, y_1, x_2, y_2 \langle / \text{box} \rangle$.”
	User:	“Where is the location of the [target]?”
	StateVLM:	“response: Here is the location of the [target]. $\langle \text{box} \rangle x_1, y_1, x_2, y_2 \langle / \text{box} \rangle$.”
Affordance reasoning (object grasp prediction)	User:	“Hand me the [target].”
	StateVLM:	“response: Sure. I will hand you the [target]. $\langle \text{box} \rangle x_1, y_1, x_2, y_2 \langle / \text{box} \rangle$.”
	User:	“Pass me the [target].”
	StateVLM:	“response: Alright, I will pick up the [target] for you. $\langle \text{box} \rangle x_1, y_1, x_2, y_2 \langle / \text{box} \rangle$.”
	User:	“Give me the [target].”
	StateVLM:	“response: Okay, I will give you the [target]. $\langle \text{box} \rangle x_1, y_1, x_2, y_2 \langle / \text{box} \rangle$.”

3.2.2. Annotation Rules

When assessing the state of kitchen tools, multiple dimensions must be considered, including contents, usage, hygiene, physical condition, position, accessibility, and operational status. This dataset challenge specifically focuses on **clearing the table** after a meal. Our scope within this task is defined as follows: containers are analyzed based on their contents, specifically whether they are empty or contain food, and what type of food they contain. Cutlery and napkins are evaluated based on hygiene and readiness for usage; they are categorized simply as clean or dirty. For object-state localization, we generated two distinct expressions for each object based on the ambiguous opposites and synonyms listed in Table 2. Overall, we obtained 25,401 expressions and 76,203 dialog instances.

3.2.3. Evaluation Metric

Our main challenges lie in accurately localizing and predicting bounding boxes for object states and grasp regions. Following prior studies [16, 12, 7, 14], we adopt the standard Intersection Over Union (IoU) [45] as our evaluation metric. The standard IoU is a widely used measure in computer vision for tasks such as object detection and segmentation. Let $B_p \subset \mathbb{R}^2$ denote the predicted bounding box and $B_g \subset \mathbb{R}^2$ the corresponding ground-

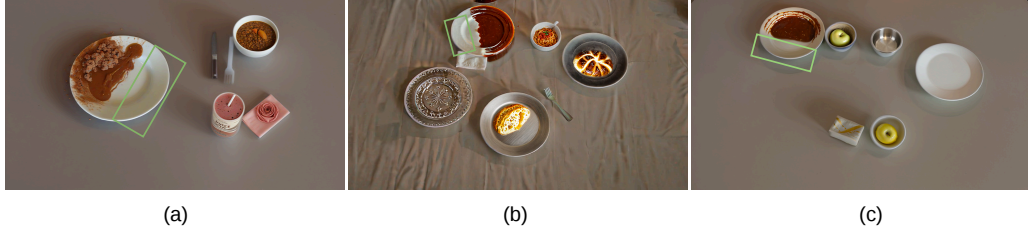


Figure 3: Example scenes in OSAR: (a) **Simple scenes** are defined as those with only one object from each category. (b) and (c) **Complex scenes** are defined as those with multiple objects from each category in various states. The red box marks the ideal grasp region; this is an example of how an object’s state affects its affordance, as the area covered with food should be avoided when grasping.

Table 2: Ambiguous opposites and synonyms in OSAR: Tableware states reflect the shifting semantics of use and cleanliness.

Object Category	Physical Condition	Semantic Interpretation
Plate, Bowl Cup, Mug, Glass	Empty, no residue	clean / unused
	Empty with little residue	used / dirty
	With / holding contents	used / dirty
Bottle	Cap open	possibly used (depends on liquid level)
	Cap closed / unopened	possibly unused (depends on liquid level)
Spoon, Fork, Knife	No residue	clean / unused
	Has residue	dirty / used
Napkin	Has residue	used / dirty
	Unfolded	
	No residue Folded neatly	clean / unused

truth box. The intersection and union are defined as

$$I = B_p \cap B_g, \quad U = B_p \cup B_g.$$

The standard IoU is

$$\text{IoU}(B_p, B_g) = \frac{|I|}{|U|},$$

where $|\cdot|$ denotes the area measure. Standard IoU is bounded in $[0, 1]$.

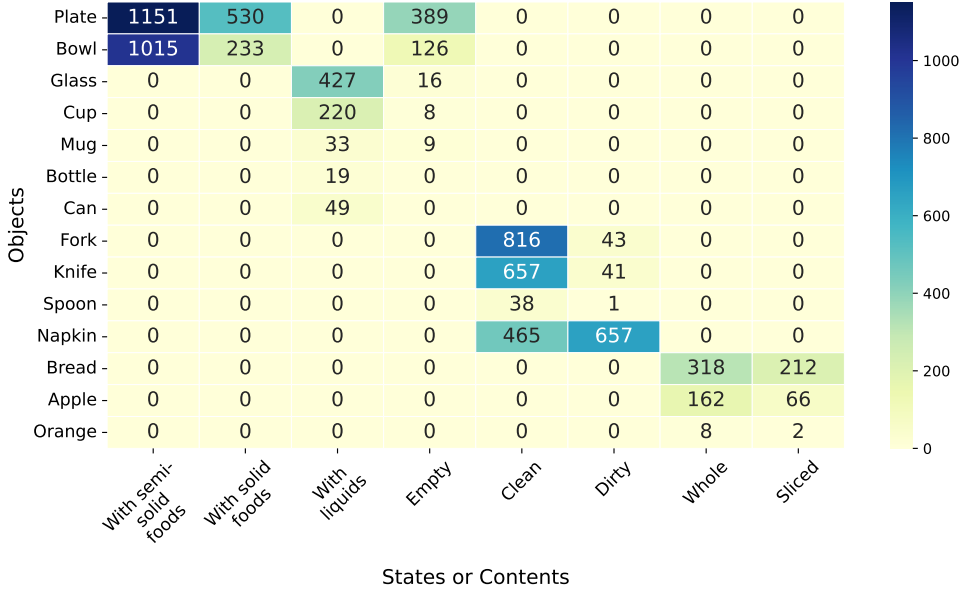


Figure 4: Object statistics in OSAR. Semi-solid foods usually include sauces, pasta, soup, noodles, and creams. Solid foods typically refer to various states of apples, bread, and oranges. Liquids encompass a variety of drinks, beverages, and juices.

4. Proposed Method: StateVLM

4.1. Overall Structure

We propose StateVLM, a state-aware vision-language model that uses MiniCPM-V [46] as its backbone. MiniCPM-V is designed for edge devices and offers strong commonsense reasoning capabilities, which are essential for robotic tasks. However, it lacks inherent object detection functionality. These characteristics make it an ideal testbed for isolating and validating our hypothesis regarding object, object-state localization and affordance reasoning. We propose a less intrusive strategy than Pix2Emb, NExT-Chat [16], to fine-tune StateVLM. Specifically, during the training phase, StateVLM utilizes the output of the box decoder to compute an auxiliary regression loss for detection training, as shown in Fig. 1a. During the inference phase, StateVLM continues to use the standard Pix2Seq format, as shown in Fig. 1b.

4.2. StateVLM Modules

4.2.1. BackBone

MiniCPM-V 2.6 [46] is adopted as the backbone network, which comprises three key modules: a visual decoder, a compression layer, and an LLM. The visual decoder leverages an adaptive visual encoding approach, SigLip-400M [47], to tokenize the image inputs into visual tokens. The compression layer uses a perceiver-resampler structure [9] with a single cross-attention layer to compress the large number of visual tokens into a fixed set of 96 tokens. Finally, the compressed visual tokens and the text input are fed into the LLM, a causal language model (Qwen2-7B [48]), which uses a specific linear head to predict the next word.

4.2.2. Box Decoder

Building on the backbone network, we propose a box decoder module, a lightweight feed-forward linear head to transform the sequence output into continuous values. Specifically, we first perform global average pooling over the temporal (sequence) dimension using an `AdaptiveAvgPool1d` layer, reducing each hidden-state sequence to a single vector. The resulting vector is flattened and passed through a two-layer MLP with a ReLU non-linearity. The MLP maps the LLM hidden size to a lower-dimensional intermediate representation and then to a 4-dimensional output corresponding to the four continuous numbers, which are the predicted location coordinates $[x_1, y_1, x_2, y_2]$.

4.3. Training Objectives

The overall training objective of StateVLM is a weighted combination of the *causal language modeling (CLM)* loss and *auxiliary regression loss (ARL)*:

$$\mathcal{L}_{\text{CLM}+\text{ARL}} = \alpha \mathcal{L}_{\text{CLM}} + \beta \mathcal{L}_{\text{ARL}},$$

$\alpha = 0.2$ and $\beta = 0.8$ are the weights of $\mathcal{L}_{\text{CLM}}$ and $\mathcal{L}_{\text{ARL}}$, which balance these two losses and ensure they play equivalent feedback roles in model tuning.

4.3.1. Causal Language Modeling

The backbone, Qwen2-7B, is trained autoregressively using the CLM objective, predicting each token conditioned on all previous tokens via a causal self-attention mask. Given a token sequence $[x_1, \dots, x_T]$ and vocabulary size V , with model logits $z_t \in \mathbb{R}^V$ at timestep t , the *CLM loss* is

$$\mathcal{L}_{\text{CLM}} = - \sum_{t=1}^{T-1} \log \frac{\exp(z_{t,x_{t+1}})}{\sum_{v=1}^V \exp(z_{t,v})},$$

where $z_{t,v}$ is the predicted logit for the token v and x_{t+1} is the corresponding ground-truth token.

4.3.2. Auxiliary Regression Loss

To enhance numerical reasoning and improve bounding box prediction, we incorporate an *ARL*. This loss jointly supervises both the location and shape of predicted bounding boxes by combining *Least Absolute Deviations (L1) loss* [49] and *Generalized Intersection Over Union (GIoU) loss* [45].

Let $B_p \subset \mathbb{R}^2$ denote the predicted bounding box and $B_g \subset \mathbb{R}^2$ the corresponding ground-truth box. The *L1 loss* is computed as

$$\mathcal{L}_{L1} = \|B_p - B_g\|_1.$$

The *GIoU loss* is based on the standard Intersection over Union (IoU) metric, which measures the overlap between two bounding boxes, as we discussed in Section 3.2.3. Because IoU does not capture spatial discrepancies when the boxes do not overlap, GIoU introduces an additional penalty term that accounts for the normalized area outside the union of the two bounding boxes but within their smallest enclosing box. Let C denote the smallest enclosing box that contains both B_p and B_g . The GIoU is defined as

$$\text{GIoU}(B_p, B_g) = \text{IoU}(B_p, B_g) - \frac{|C| - |B_p \cup B_g|}{|C|},$$

GIoU loss is calculated as

$$\mathcal{L}_{\text{GIoU}} = 1 - \text{GIoU}(B_p, B_g).$$

The total bounding box loss is a weighted sum of the two components:

$$\mathcal{L}_{\text{ARL}} = \gamma \mathcal{L}_{L1} + \delta \mathcal{L}_{\text{GIoU}},$$

$\gamma = 0.2$ and $\delta = 0.8$ follows the ratio utilized in DETR [50] and NExT-Chat [16].

5. Experiments

5.1. Experimental Overview

We conduct two experiments. The first experiment is conducted on large-scale adapted referring expression comprehension (REC) benchmarks, RefCOCO, RefCOCO+, and RefCOCOg, to explore our earlier hypothesis that the current training paradigm for numerical tasks may lead to inefficient learning. The second experiment is performed on the proposed benchmark OSAR, to investigate whether a well-annotated, small-scale dataset can improve the potential object-state awareness of VLMs. We evaluate the accuracy of predicted bounding boxes using the IoU metric (see Section 3.2.3) with a threshold of 0.5, denoted as Acc@0.5.

5.2. Implementation Details

We implement StateVLM using PyTorch² and the HuggingFace libraries³. The model is initialized with weights from the pretrained MiniCPM-V 2.6. We perform full fine-tuning of the entire model in the first experiment on large-scale public benchmarks. For the second experiment, we modify the model for parameter-efficient fine-tuning using LoRA [51] on our proposed small-scale dataset. All experiments are conducted on four NVIDIA A100 GPUs (80GB each). We employ DeepSpeed⁴ to optimize distributed training and use an automatically warmed-up learning rate of 1e-6 for both training regimes. The batch size is set to 24 for full fine-tuning and 48 for LoRA fine-tuning.

5.3. Experiment 1: Full Fine-Tuning on Adapted Public Benchmarks

To verify our hypothesis that the current training paradigm for numerical tasks may lead to inefficient learning, we conduct comparative experiments on the adapted REC benchmarks: RefCOCO, RefCOCO+ [4], and RefCOCOg [5]. RefCOCO and RefCOCO+ contain train, validation, testA, and testB splits, while RefCOCOg contains train, validation, and test splits. We fine-tune our models on the training split and report performance on the corresponding validation and test sets (i.e., testA and testB for RefCOCO and RefCOCO+, and the test set for RefCOCOg).

²<https://pytorch.org/>

³<https://huggingface.co/>

⁴<https://www.deepspeed.ai/>

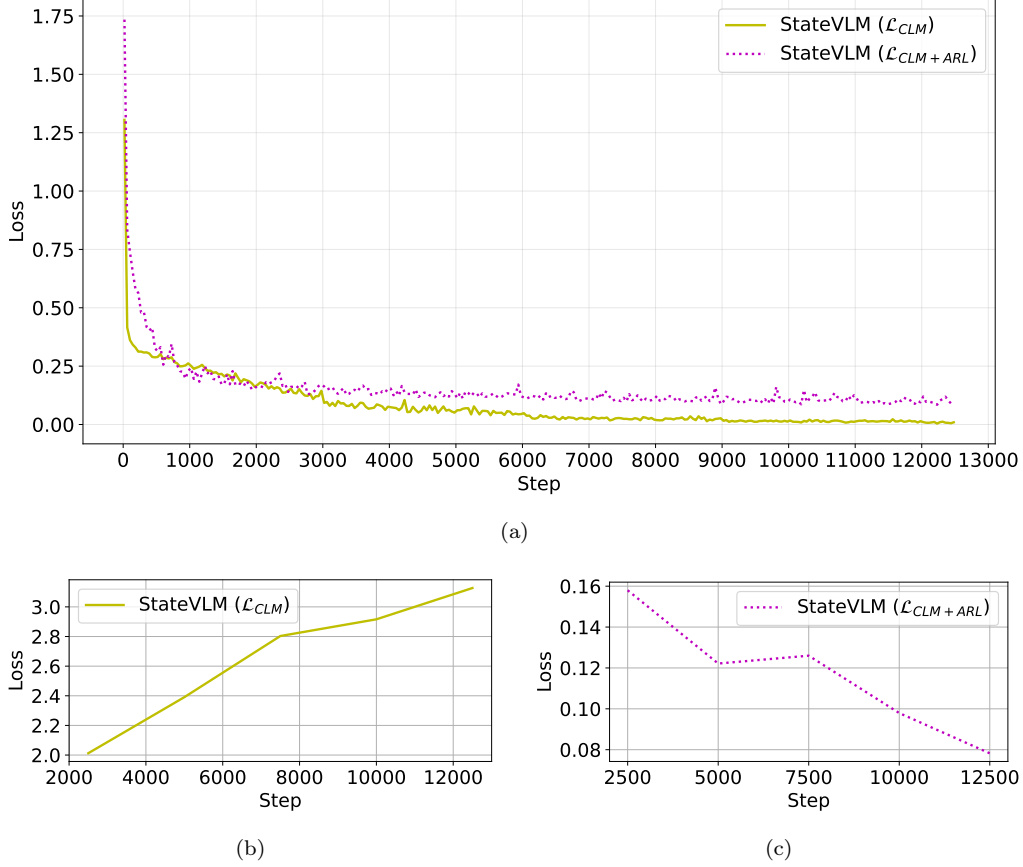


Figure 5: StateVLM loss curves: (a) StateVLM ($\mathcal{L}_{CLM}$ and $\mathcal{L}_{CLM+ARL}$) training loss progression over steps, (b) Validation loss for StateVLM ($\mathcal{L}_{CLM}$), and (c) Validation loss for StateVLM ($\mathcal{L}_{CLM+ARL}$).

We first assess the performance of the backbone model, MiniCPM-V, as a baseline. Then, we conduct two groups of full fine-tuning experiments: one in which the model is trained solely with the standard *CLM* objective and another in which it is trained using the *CLM* combined with the *ARL* objective described in Section 4.3. For the distinction, we denote them as StateVLM ($\mathcal{L}_{CLM}$) and StateVLM ($\mathcal{L}_{CLM+ARL}$), respectively.

5.3.1. Results and Analysis

We observe that the training loss varies with the number of steps for each model. StateVLM ($\mathcal{L}_{CLM}$) and StateVLM ($\mathcal{L}_{CLM+ARL}$) show a similar trend, where the losses gradually decrease as the number of training steps increases,

as shown in Fig. 5a. The validation loss for StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) aligns with the training loss trend (see Fig. 5c). However, the validation loss for StateVLM ($\mathcal{L}_{\text{CLM}}$) continues to increase with the number of training steps (see Fig. 5b). StateVLM ($\mathcal{L}_{\text{CLM}}$) with 5000 training steps outperforms both the 2500-step and 7500-step models. We suspect that StateVLM ($\mathcal{L}_{\text{CLM}}$) begins to overfit after 5000 steps of training on the current training data.

Furthermore, we evaluate the fine-tuned models StateVLM ($\mathcal{L}_{\text{CLM}}$) and StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) on the validation and test splits of RefCOCO, RefCOCO+, and RefCOCOg. We used a fixed random seed for evaluation to ensure deterministic model outputs. Consequently, all runs yield identical results, and no deviation metrics are applicable. We demonstrate the average performance of the two models at 5,000, 10,000, and 15,000 training steps on the RefCOCO, RefCOCO+, and RefCOCOg benchmarks in Fig. 6. StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) performance improves significantly from 5,000 to 15,000 steps and is better than StateVLM ($\mathcal{L}_{\text{CLM}}$) after 10,000 steps. StateVLM ($\mathcal{L}_{\text{CLM}}$) performance, however, remains stable or even decreases over the training, which is consistent with the validation loss curve in Fig. 5b.

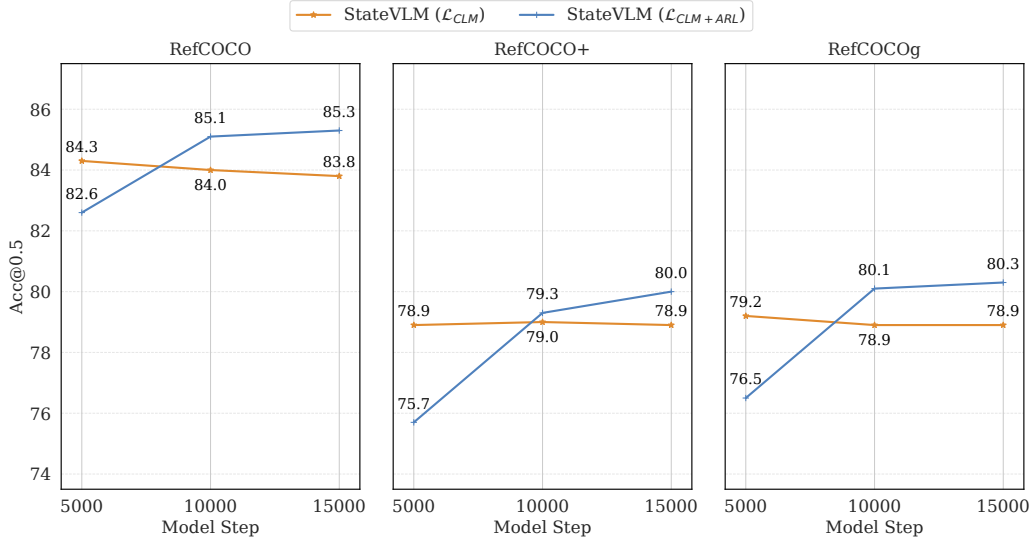


Figure 6: Average performance of StateVLM ($\mathcal{L}_{\text{CLM}}$) and StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) across benchmarks over training Steps.

The concrete performance of StateVLM ($\mathcal{L}_{\text{CLM}}$) on the subset of RefCOCO, RefCOCO+, and RefCOCOg splits is illustrated in Fig. 7. The

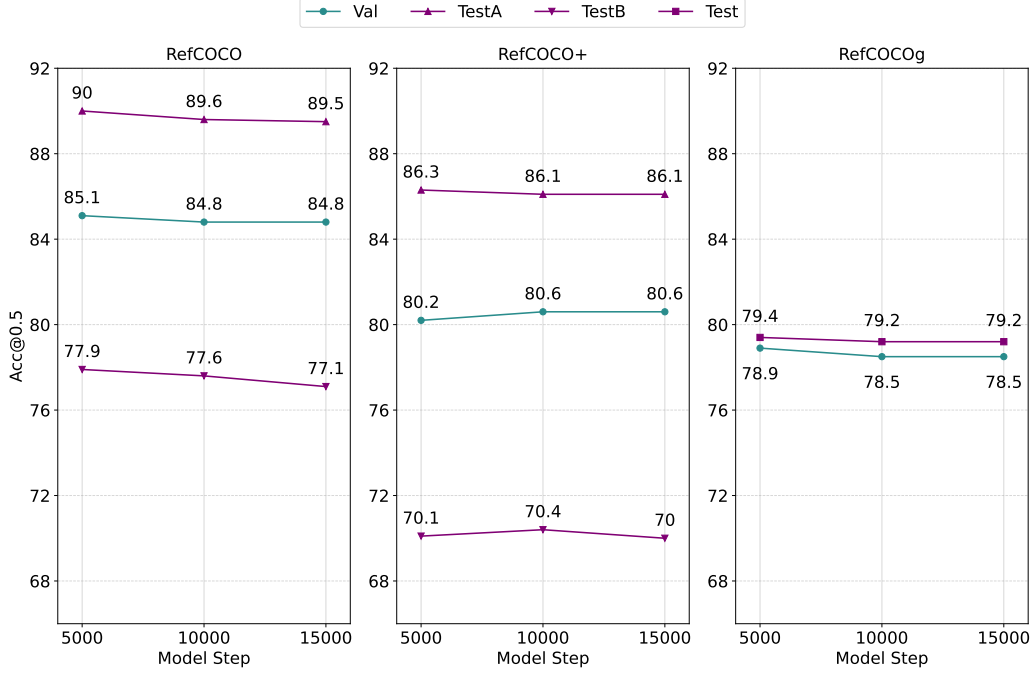


Figure 7: StateVLM ($\mathcal{L}_{\text{CLM}}$) performance changes over training steps.

performance of StateVLM ($\mathcal{L}_{\text{CLM}}$) remains stable between 5,000 and 10,000 steps and decreases at 15,000 steps, indicating that it reaches its maximum performance at 5,000 steps. We continue training the StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) until 25,000 steps and the performance gradually improves over the training steps on RefCOCO, RefCOCO+, and RefCOCOg, as shown in Fig. 8.

The performance of StateVLM ($\mathcal{L}_{\text{CLM}}$) and StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) is illustrated in Table 3, alongside existing models on the adapted REC task. StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) outperforms StateVLM ($\mathcal{L}_{\text{CLM}}$), demonstrating the effectiveness of the auxiliary regression loss in improving the model’s performance on bounding box prediction tasks. Compared with the Pix2Emb-based method NExT-Chat, StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) achieves superior performance. Comparing the Pix2Seq-based method StateVLM ($\mathcal{L}_{\text{CLM}}$) with Shikra, StateVLM ($\mathcal{L}_{\text{CLM}}$) achieves lower performance than Shikra. As the training recipe, computing resources, and training time vary, we cannot directly infer why StateVLM ($\mathcal{L}_{\text{CLM}}$) underperforms Shikra. However, StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) achieves performance comparable to Shikra, demonstrating the effectiveness of the auxiliary regression loss. On average, the Stat-

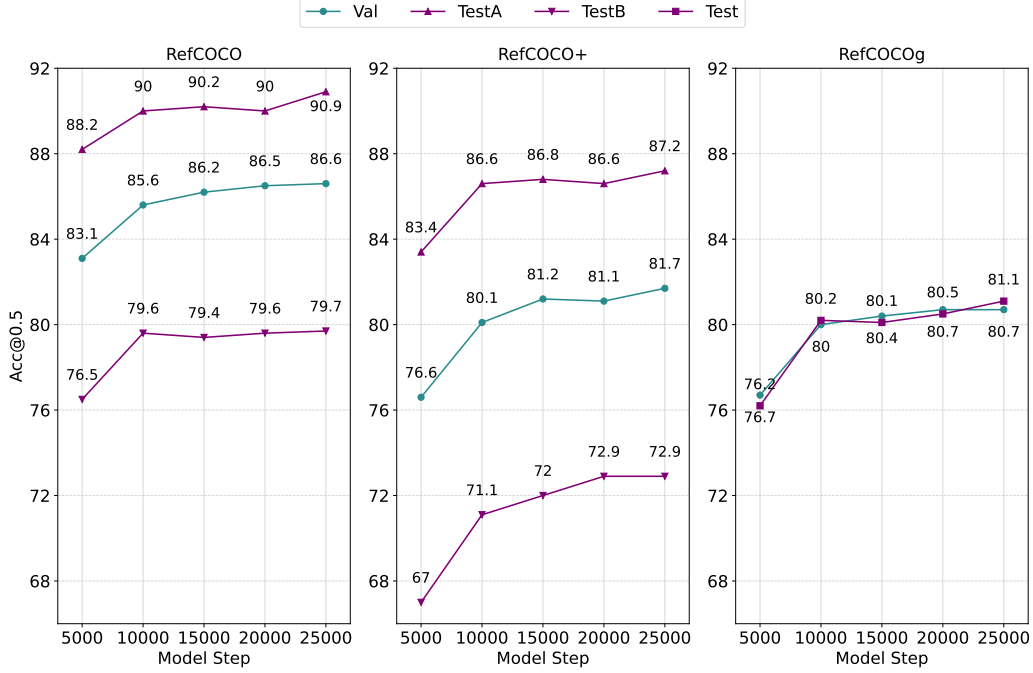


Figure 8: StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) performance changes over training steps.

eVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) achieves a 1.6% performance improvement over the StateVLM ($\mathcal{L}_{\text{CLM}}$). Overall, these results demonstrate that incorporating an auxiliary regression loss can significantly enhance the performance of VLMs on bounding box prediction tasks, validating our hypothesis about the limitations of the current training paradigm for object localization tasks.

Ablation Study. We conducted several ablation studies to identify the components essential for achieving the final performance.

Box Decoder Design: One way is to reduce the two linear layers to a single linear layer. Additionally, we tried different activation functions within the MLP, testing GELU and Sigmoid.

Text Prompts: We tested two different text prompts to understand the effect of text content on model performance:

- Simple Prompt: “A chat between a human and an AI that understands visuals. Follow instructions.”
- Concrete Prompt: “A chat between a human and an AI that understands visuals. In images, [x, y] denotes points: top-left is [0, 0],

Table 3: Comparative performance comparison in adapted REC benchmarks. The evaluation metric is Acc@0.5.

VLMs Type	Methods	RefCOCO			RefCOCO+			RefCOCOg	
		val	testA	testB	val	testA	testB	val	test
Pix2Emb-7B	NExT-Chat [16]	85.5	90.0	77.9	77.2	84.5	68.0	80.1	79.8
	Shikra [7]	87.0	90.6	80.2	81.6	87.4	72.1	82.3	82.2
Pix2Seq-7B	Baseline (MiniCPM-V)	16.2	19.6	13.4	12.4	15.7	9.9	7.2	6.7
	StateVLM ($\mathcal{L}_{\text{CLM}}$)	85.1	90.0	77.9	80.2	86.3	70.1	78.9	79.4
	StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$)	86.6	90.9	79.7	81.7	87.2	72.9	80.7	81.1

bottom-right is $[1, 1]$. Increasing x moves right; y moves down. A bounding box is defined as $[x_1, y_1, x_2, y_2]$ where (x_1, y_1) is the top-left corner and (x_2, y_2) is the bottom-right corner. Follow instructions.”

Overall, the combination of the MLP architecture that we presented, along with the simple prompt, achieved the best performance.

5.4. Experiment 2: LoRA Fine-Tuning on Proprietary Dataset

To investigate the potential object-state awareness of VLMs, we introduce a small-scale and well-annotated dataset, OSAR, as detailed in Section 3. We first evaluate the performance of the baseline model, MiniCPM-V, and the models that we obtained from previous experiments: StateVLM ($\mathcal{L}_{\text{CLM}}$) and StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) on our proposed dataset. Due to the cost of dataset generation and annotation, OSAR is relatively small-scale, which is not sufficient for full fine-tuning. Therefore, we further fine-tune these models by the LoRA strategy, obtaining the models StateVLM ($\mathcal{L}_{\text{CLM}}$, LoRA) and StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$, LoRA).

5.4.1. Results and Analysis

A systematic experimental evaluation on the proposed tasks: object detection and affordance reasoning is shown in Fig. 9. The baseline model, MiniCPM-V, fails to perform these tasks, as it is not trained for object detection. StateVLM ($\mathcal{L}_{\text{CLM}}$) that we obtained from the previous experiment is able to perform on our proposed benchmark. It performs poorly than StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$), achieving scores of 34.3%. StateVLM ($\mathcal{L}_{\text{CLM}}$, LoRA) that LoRA fine-tuned on our dataset improve to 55.1%, which is lower than StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$, LoRA). StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) achieve scores

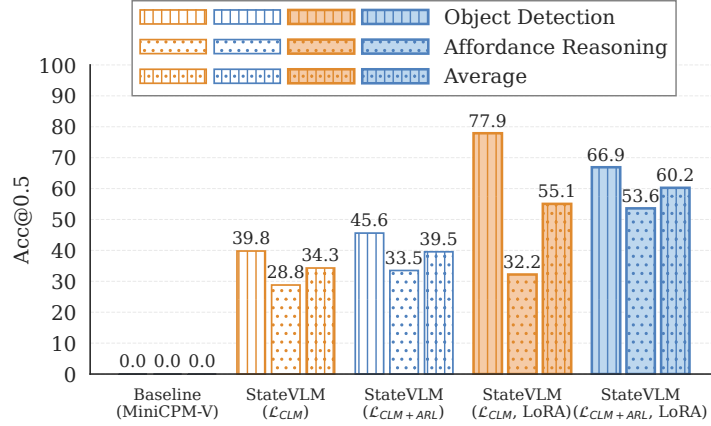


Figure 9: Comprehensive performance comparison on OSAR.

Table 4: Exception rates(%). N/A indicates Not Applicable because LoRA fine-tuning was evaluated only on OSAR, not on the adapted REC datasets.

Model	Adapted REC Datasets			Proprietary Dataset	
	RefCOCO	RefCOCO+	RefCOCOg	Object Detection	Affordance Reasoning
StateVLM ($\mathcal{L}_{CLM}$)	0.21	0.15	0.17	0.10	0.06
StateVLM ($\mathcal{L}_{CLM+ARL}$)	0.00	0.00	0.00	0.00	0.00
StateVLM ($\mathcal{L}_{CLM}$, LoRA)	N/A	N/A	N/A	3.45	33.36
StateVLM ($\mathcal{L}_{CLM+ARL}$, LoRA)	N/A	N/A	N/A	0.00	0.00

of 39.5%. StateVLM ($\mathcal{L}_{CLM+ARL}$, LoRA) increases to 60.2%. These results demonstrate that the auxiliary regression loss is effective in improving the model’s performance on our proposed benchmark, which is consistent with the conclusion of the first experiment. Notably, the performance of StateVLM ($\mathcal{L}_{CLM+ARL}$, LoRA) on the object detection task is lower than that of StateVLM ($\mathcal{L}_{CLM}$, LoRA), which is inconsistent with the conclusion of the first experiment. However, StateVLM ($\mathcal{L}_{CLM+ARL}$, LoRA) performs better than StateVLM ($\mathcal{L}_{CLM}$, LoRA) on the affordance reasoning task, which is consistent with the conclusion of the first experiment.

The affordance reasoning task is more complex and challenging than the object detection task. The object detection task only requires the model to predict the bounding box of the object, whereas the affordance reasoning task requires the model to predict the bounding box of the grasping area, which is sometimes only a part of the object. In previous full fine-tuning benchmarks (RefCOCO, RefCOCO+, and RefCOCOg), the bounding boxes generally enclose the whole object rather than just a part of it. Therefore, StateVLM

($\mathcal{L}_{\text{CLM}}$, LoRA) quickly learns to predict the bounding box of the whole object, improving from 39.8% to 77.9% after LoRA fine-tuning. However, it is limited to learn to predict the bounding box of the grasping area, improving only from 28.8% to 32.2% after LoRA fine-tuning. It also exhibits a high exception rate of 33.36% on the affordance reasoning task, indicating that the model does not fully understand the complexity of the task and to predict bounding boxes in a consistent format.

In contrast, StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$, LoRA) is able to learn to predict both the bounding box of the whole object and the grasping area simultaneously. It achieves a zero exception rate on both tasks, indicating that the model learns to understand the tasks and to produce bounding boxes in a consistent format. We conclude that the auxiliary regression loss enhances the model’s ability to predict bounding boxes in a consistent, predefined format, which is particularly beneficial for the affordance reasoning task. The exception rates of all models on the subtasks are shown in Table 4. Models trained with the auxiliary regression loss achieve zero exception rates, further demonstrating its effectiveness in improving model performance and output consistency on numerical tasks.

5.4.2. Visualization

To provide a more direct and intuitive observation of the performance of StateVLM ($\mathcal{L}_{\text{CLM}}$) and StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$), we visualize representative successful and failed instances of predicted bounding boxes from these two models on object detection and affordance reasoning tasks, as shown in Fig. 10. For object detection, the goal is to predict bounding boxes covering entire objects. Both models generally succeed on this task but exhibit reduced accuracy on cutlery (e.g., knives, forks, and spoons). The predicted boxes for these cases are often biased toward one side of the object. This suggests a potential interference from affordance reasoning, as cutlery also represents the most challenging category for that task. For affordance reasoning, the objective is to localize grasping regions. StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) demonstrates improved understanding of affordances for liquids and semi-solid foods (e.g., sauces and drinks), accurately predicting grasping regions for their containers. In contrast, StateVLM ($\mathcal{L}_{\text{CLM}}$) frequently produces invalid prediction formats or does not capture the intended task. Both models struggle with plates and cutlery, failing to correctly localize their grasping regions. Overall, incorporating the auxiliary regression loss improves affordance reasoning while a significant gap remains between current performance

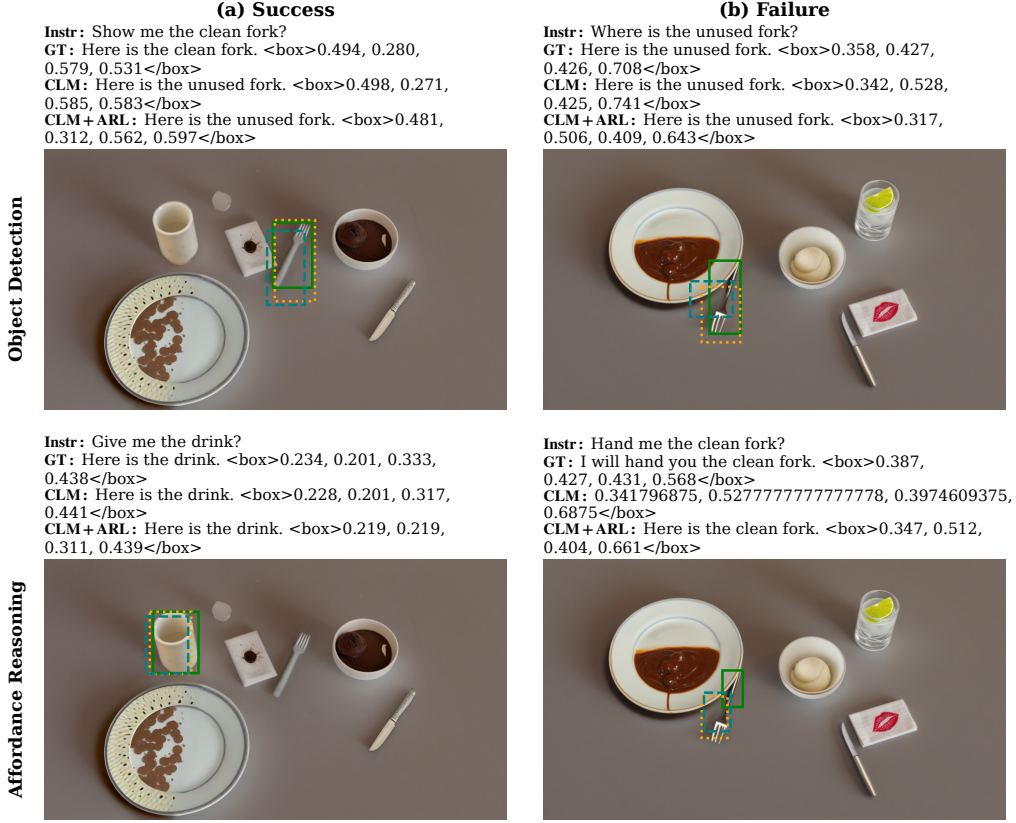


Figure 10: Qualitative comparison of StateVLM ($\mathcal{L}_{\text{CLM}}$) and StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$) on grounded object detection and affordance reasoning examples. Ground-truth boxes are shown in green, while model predictions are shown in orange (StateVLM ($\mathcal{L}_{\text{CLM}}$)) and teal (StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$)).

and practical deployment, motivating further research.

6. Conclusion and Future Work

In this paper, we propose StateVLM, a novel model designed to perceive and learn fine-grained object representations, including precise localization of objects and their states, as well as graspable regions. We propose a less intrusive strategy than Pix2Emb for adapting StateVLM to numerical tasks under limited computational resources. Specifically, during the training phase, StateVLM utilizes the output of the box decoder to compute an auxiliary regression loss (ARL) for detection training. During the inference

phase, StateVLM continues with standard sequence prediction. Comparative results on adapted REC tasks (RefCOCO, RefCOCO+, RefCOCOg) demonstrate that this ARL improves the performance of StateVLM on REC tasks compared to training without it, thereby validating our hypothesis about the limitations of the current training paradigm of Pix2Seq for numerical tasks. StateVLM’s performance on the proposed OSAR benchmark indicates that its latent state-aware capabilities for object-state localization and affordance reasoning can be further enhanced through LoRA tuning on a small-scale and well-annotated dataset. Additionally, we find that this ARL also enhances the consistency of StateVLM outputs, which is particularly important for complex tasks such as affordance reasoning.

There remain challenges to our proposed approach for real-world embodied agents. First, affordance is a broad and multifaceted concept, and we hypothesize that more extensive and carefully annotated datasets are required to advance this field. Although we leveraged a diffusion model to generate the dataset, human verification is still necessary. Second, our evaluation focused on distinguishing objects in different states, and we did not investigate ambiguous or vague instructions in depth. Addressing these challenges will require the development of novel approaches that extend beyond efficient fine-tuning strategies. However, we demonstrate that incorporating an ARL significantly improves StateVLM performance in object detection, and that a small-scale, well-annotated dataset further enhances StateVLM’s state-awareness, thereby improving object-state localization and affordance reasoning. With the publication of this research, we will release the code and dataset to facilitate further research in this area.

7. Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

8. Acknowledgments

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Appendix A. Appendix

We summarize the configurations of VLMs capable of performing the REC task in Table A.5, including their architectures, names, sizes, training resources, and training stages. Table A.6 further presents the performance of these VLMs on adapted REC benchmarks, including RefCOCO, RefCOCO+, and RefCOCOg. Due to differences in the amount of training resources, training strategies, and data diversity, the performance of these VLMs varies significantly across benchmarks. Our goal is not to achieve state-of-the-art performance on these benchmarks, but rather to verify the effectiveness of the auxiliary regression loss in improving VLM performance on numerical tasks, particularly object localization.

Table A.5: Summary of VLM configurations capable of performing the REC task.
 * refers to different resources to be used in the second stage. × states that this stage is not included. Pix2Seq and Pix2Emb stand for pixel-to-sequence and pixel-to-embedding.

VLMs Architecture	VLMs Name	VLMs Size	Training GPUs	Training Stage 1	Training Stage 2
Pix2Seq	Shikra [7]	~13B	8×A100	~100 hours	~20 hours
	Ferret [12]		8×A100	~125 hours	×
	Ferret (v2) [13]		8×A100	N/A	N/A
	SPHINX-1K [6]		32×A100	~125 hours	~38 hours*
	SPHINX-2K [6]		32×A100	~250 hours	~38 hours*
	VistaLLM [11]		32×A100	~72 hours	~30 hours
	CogVLM-G [14]		256×A100	~120,000 steps	~60,000 steps
	Shikra [7]	~7B	8×A100	~100 hours	~20 hours
	MiniGPT (v2) [8]		8×A100	~90 hours	~20 hours*
	Qwen-VL [9]		N/A	~50,000 steps	~19,000 steps
	LLaVA-G [10]		N/A	~10,000 steps	~8,000 steps
	VistaLLM [11]		32×A100	~48 hours	~22 hours
	Ferret [12]		8×A100	~60 hours	×
	Ferret (v2) [13]		8×A100	N/A	N/A
Pix2Emb	NExT-Chat [16]	~7B	8×A100	~59 hours	~10 hours

Table A.6: StateVLM: Comparative performance comparison in REC. The evaluation metric is Acc@0.5. * refers to the specialist or fine-tuned methods.

VLMs Type	Methods	RefCOCO			RefCOCO+			RefCOCOg	
		val	testA	testB	val	testA	testB	val	test
non-VLM	MAttNet* [52]	76.4	80.4	69.3	64.9	70.3	56.0	66.7	67.0
	OFA-L [27]	80.0	83.7	76.4	68.3	76.0	61.8	67.6	67.6
	TransVG* [28]	81.0	82.7	78.4	64.8	70.7	56.9	68.7	67.7
	UNITER* [29]	81.4	87.0	74.2	75.9	81.5	66.7	74.0	68.7
	VILLA* [30]	82.4	87.5	74.8	76.2	81.5	66.8	76.2	76.7
	UniTAB* [31]	86.3	88.8	80.6	78.7	83.2	69.5	80.0	80.0
	G-DINO-L* [32]	90.6	93.2	88.2	82.8	89.0	75.9	86.1	87.0
Pix2Seq-13B	Shikra [7]	87.8	91.1	81.8	82.9	87.8	74.4	82.6	83.2
	Ferret [12]	89.5	92.4	84.4	82.8	88.1	75.2	85.8	86.3
	Ferret (v2) [13]	92.6	95.0	88.9	87.4	92.1	81.4	89.4	90.0
	SPHINX [6]	89.2	91.4	85.1	82.8	87.3	76.9	84.9	83.7
	SPHINX-1K [6]	91.1	92.7	86.7	86.6	91.1	80.4	88.2	88.4
	SPHINX-2k [6]	91.1	92.9	87.1	85.5	90.6	80.5	88.1	88.7
	VistaLLM [11]	89.9	92.5	85.0	84.1	90.3	75.8	86.0	86.4
	CogVLM-G [14]	92.8	94.8	89.0	88.7	93.0	83.4	89.8	90.8
Pix2Seq-7B	Shikra [7]	87.0	90.6	80.2	81.6	87.4	72.1	82.3	82.2
	MiniGPT (v2) [8]	88.1	91.3	84.3	79.6	85.5	73.3	84.2	84.31
	Qwen-VL [9]	88.6	92.3	84.5	82.8	88.6	76.8	86.0	86.3
	LLaVA-G [10]	89.2	—	—	81.7	—	—	84.8	—
	VistaLLM [11]	88.1	91.5	83.0	82.9	89.8	74.8	83.6	84.4
	Ferret [12]	87.5	91.4	82.5	80.9	87.4	73.1	83.9	84.8
	Ferret (v2) [13]	92.8	94.7	88.7	87.4	92.8	79.3	89.4	89.3
Pix2Emb-7B	NExT-Chat [16]	85.5	90.0	77.9	77.2	84.5	68.0	80.1	79.8
Pix2Seq-7B	Baseline (MiniCPM-V)	16.2	19.6	13.4	12.4	15.7	9.9	7.2	6.7
	StateVLM ($\mathcal{L}_{\text{CLM}}$)	85.1	90.0	77.9	80.2	86.3	70.1	78.9	79.4
	StateVLM ($\mathcal{L}_{\text{CLM}+\text{ARL}}$)	86.6	90.9	79.7	81.7	87.2	72.9	80.7	81.1